

11/12/25

Task-8 - Implement various txt/csv file operations

8.1 Aim:

To write a python program for creating and updating student registration details using txt file operations.

Algorithm:

Step:1 - Start

Step:2 - Using open() method. (create and write text files "myfile.txt" with student details)

Step:3 - Open the file in read mode and using read() method print the student details

Step:4 - Using seek method print the particular student record

Step:5 - Using tell method print the current position of the file

Step:6 - Close the file

Step:7 - Stop

Program:

```
file = open("student1.txt", "w")
```

```
input1 = input("Enter columns name\n")
```

```
file.write(input1)
```

```
file.write("\n")
```

```
n = int(input("Enter the no of students"))
```

```
for i in range(0, n):
```

```
    input2 = input("Enter student details with fr new")
```

```
    file.write(input2)
```

```
    file.write("\n")
```

Output:

Student Details using Read function is:

VTUNo	Name	Age
-------	------	-----

2305	Ram	20
------	-----	----

1920	Shiva	21
------	-------	----

2305	Ram	20
------	-----	----

1920	Shiva	21
------	-------	----

The length of first line is:

15

Output of Readline (first student record) function is:

2305 Ram 20

Find the current position of file pointer:

29

Output:

Uppercase letters: 5

Lowercase letters: 48

Digits: 7

5, 48, 4

```
file = open("student1.txt", "a")
input3 = input("Enter updated student details\n")
file.write(input3)
file = open("student1.txt", "r")
print("Student details using Read function is:")
print(file.read())
print("\n")
file.seek(0)
print("The length of first line is:")
line = file.readline()
len = len(line) (line)
print(len)
file.seek(len+1)
print("Output of Read line (first student record) function is:")
print(file.readline())
print("\n find the current position of file pointer:")
f = file.tell()
print(f)
file.close()
```

B.2 Counting cases (lowercase & upper case)

Input:

File name: merge.txt

Output: 5, 48, 4

Program:

with open("merge.txt", "w") as f:

f.write("Python is a High level language developed

by Guido Van Rossum in 1991")

```
with open("merge.txt", "r") as f:
```

```
    text = f.read()
```

```
    upper_count = 0
```

```
    lower_count = 0
```

```
    digit_count = 0
```

```
    for char in text:
```

```
        if char.isupper():
```

```
            upper_count += 1
```

```
        elif char.islower():
```

```
            lower_count += 1
```

```
        elif char.isdigit():
```

```
            digit_count += 1
```

```
    print("Uppercase letters:", upper_count)
```

```
    print("Lowercase letters:", lower_count)
```

```
    print("Digits:", digit_count)
```

```
    print(f"{upper_count}, {lower_count}, {digit_count}")
```

B.3 Construct a python program to read the above text of students grades from a text file (grades.txt) calculate average grade for each student and print out the result as students name along with their average grade using another text file.

Program:

```
with open("grades.txt", "r") as f:
```

```
    lines = f.readlines()
```

```
    n = int(lines[0].strip())
```

```
    for i in range(1, n+1):
```

Output: results.txt

Gaurav --> 169.0

Abhinav --> 138.0

Harvard --> 152.0

Jai --> 163.0

Ravi --> 188.0

```
weights = [float(w) for w in weights]
```

```
students = []
```

```
for i in range(2, 2+n):
```

```
    parts = lines[i].strip().split()
```

```
    names = parts[0]
```

```
    marks = [int(m) for m in parts[1:]]
```

```
    total = 0
```

```
    for j in range(4):
```

```
        total = total + marks[j] * weights[j]
```

```
    students.append(name, round(total, 2))
```

```
with open("results.txt", "w") as f:
```

```
    for name, avg in students
```

```
        f.write(name + "-->" + str(avg) + "\n")
```

```
print("Average grade have been written to results.  
txt")
```

VEL TECH - CSE	
EX NO.	8
PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	3
VIVA VOCE (3)	3
RECORD (4)	4
TOTAL (15)	15
DATE	

Result:

Thus the python program for implementing the text file operations was executed successfully.